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In [1]: # # This mounts your Google Drive to the Colab VM.
# from google.colab import drive
# drive.mount('/content/drive')

# # TODO: Enter the foldername in your Drive where you have saved the unzipped
# # assignment folder, e.g. 'cs6353/assignments/assignment2/'
# FOLDERNAME = 'assignment2'
# assert FOLDERNAME is not None, "[!] Enter the foldername."

# # Now that we've mounted your Drive, this ensures that
# # the Python interpreter of the Colab VM can load
# # python files from within it.
# import sys
# sys.path.append('/content/drive/My Drive/{}'.format(FOLDERNAME))

# # This downloads the CIFAR-10 dataset to your Drive
# # if it doesn't already exist.
# %cd /content/drive/My\ Drive/$FOLDERNAME/cs6353/datasets/
# !bash get_datasets.sh
# %cd /content/drive/My\ Drive/$FOLDERNAME

# # Install requirements from colab_requirements.txt
# # TODO: Please change your path below to the colab_requirements.txt file
# ! python -m pip install -r /content/drive/My\ Drive/$FOLDERNAME/colab_requirement

# ----- Laptop -----
# %cd "/home/jared/Desktop/Fall2025/CS-6353/Homeworks/assignment2/cs6353/datasets/"
# !bash get_datasets.sh
# %cd "/home/jared/Desktop/Fall2025/CS-6353/Homeworks/assignment2"
# ----- Laptop -----

# ----- Desktop -----
%cd "/home/jared/CS-6353/Homeworks/assignment2/cs6353/datasets/"
!bash get_datasets.sh
%cd "/home/jared/CS-6353/Homeworks/assignment2/"
# ----- Desktop -----

# Install requirements from colab_requirements.txt
# TODO: Please change your path below to the colab_requirements.txt file

# ----- Desktop -----
! python -m pip install -r "/home/jared/CS-6353/Homeworks/assignment2/colab_require
# ----- Desktop -----

# ----- Laptop -----
# ! python -m pip install -r "/home/jared/Desktop/Fall2025/CS-6353/Homeworks/assign
# ----- Laptop -----
```

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/home/jared/CS-6353/Homeworks/assignment2/cs6353/datasets
--2025-10-01 19:44:03-- http://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
Resolving www.cs.toronto.edu (www.cs.toronto.edu)... 128.100.3.30
Connecting to www.cs.toronto.edu (www.cs.toronto.edu)|128.100.3.30|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 170498071 (163M) [application/x-gzip]
Saving to: 'cifar-10-python.tar.gz'

```

```
cifar-10-python.tar 100%[=====>] 162.60M 6.62MB/s in 24s
```

```
2025-10-01 19:44:27 (6.85 MB/s) - 'cifar-10-python.tar.gz' saved [170498071/170498071]
```

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cifar-10-batches-py/
cifar-10-batches-py/data_batch_4
cifar-10-batches-py/readme.html
cifar-10-batches-py/test_batch
cifar-10-batches-py/data_batch_3
cifar-10-batches-py/batches.meta
cifar-10-batches-py/data_batch_2
cifar-10-batches-py/data_batch_5
cifar-10-batches-py/data_batch_1
/home/jared/CS-6353/Homeworks/assignment2
Requirement already satisfied: anyio==3.7.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 1)) (3.7.1)
Requirement already satisfied: appnope==0.1.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 2)) (0.1.3)
Requirement already satisfied: argon2-cffi==23.1.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 3)) (23.1.0)
Requirement already satisfied: argon2-cffi-bindings==21.2.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 4)) (21.2.0)
Requirement already satisfied: arrow==1.2.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 5)) (1.2.3)
Requirement already satisfied: asttokens==2.2.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 6)) (2.2.1)
Requirement already satisfied: async-lru==2.0.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 7)) (2.0.4)
Requirement already satisfied: attrs==23.1.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 8)) (23.1.0)
Requirement already satisfied: Babel==2.12.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 9)) (2.12.1)
Requirement already satisfied: backcall==0.2.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 10)) (0.2.0)
Requirement already satisfied: beautifulsoup4==4.12.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 11)) (4.12.2)

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Requirement already satisfied: bleach==6.0.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 12)) (6.0.0)

Requirement already satisfied: certifi==2023.7.22 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 13)) (2023.7.22)

Requirement already satisfied: cffi==1.15.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 14)) (1.15.1)

Requirement already satisfied: charset-normalizer==3.2.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 15)) (3.2.0)

Requirement already satisfied: comm==0.1.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 16)) (0.1.4)

Requirement already satisfied: contourpy==1.1.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 17)) (1.1.0)

Requirement already satisfied: cycler==0.11.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 18)) (0.11.0)

Requirement already satisfied: debugpy==1.6.7.post1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 19)) (1.6.7.post1)

Requirement already satisfied: decorator<=5.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 20)) (4.4.2)

Requirement already satisfied: defusedxml==0.7.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 21)) (0.7.1)

Requirement already satisfied: executing==1.2.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 22)) (1.2.0)

Requirement already satisfied: fastjsonschema==2.18.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 23)) (2.18.0)

Requirement already satisfied: fonttools==4.42.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 24)) (4.42.1)

Requirement already satisfied: fqdn==1.5.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 25)) (1.5.1)

Requirement already satisfied: idna==3.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 26)) (3.4)

Requirement already satisfied: imageio==2.31.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 27)) (2.31.1)

Requirement already satisfied: ipykernel<=5.5.6 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 28)) (5.5.6)

Requirement already satisfied: ipython<=7.34.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 29)) (7.34.0)

Requirement already satisfied: isoduration==20.11.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment

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2/colab_requirements.txt (line 30)) (20.11.0)
Requirement already satisfied: jedi==0.19.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 31)) (0.19.0)
Requirement already satisfied: Jinja2==3.1.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 32)) (3.1.2)
Requirement already satisfied: json5==0.9.14 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 33)) (0.9.14)
Requirement already satisfied: jsonpointer==2.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 34)) (2.4)
Requirement already satisfied: jsonschema==4.19.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 35)) (4.19.0)
Requirement already satisfied: jsonschema-specifications==2023.7.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 36)) (2023.7.1)
Requirement already satisfied: jupyter-events==0.7.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 37)) (0.7.0)
Requirement already satisfied: jupyter-lsp==2.2.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 38)) (2.2.0)
Requirement already satisfied: jupyter_client<8.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 39)) (7.4.9)
Requirement already satisfied: jupyter_core==5.3.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 40)) (5.3.1)
Requirement already satisfied: jupyter_server==2.7.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 41)) (2.7.2)
Requirement already satisfied: jupyter_server_terminals==0.4.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 42)) (0.4.4)
Requirement already satisfied: jupyterlab==4.0.5 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 43)) (4.0.5)
Requirement already satisfied: jupyterlab-pygments==0.2.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 44)) (0.2.2)
Requirement already satisfied: jupyterlab_server==2.24.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 45)) (2.24.0)
Requirement already satisfied: kiwisolver==1.4.5 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 46)) (1.4.5)
Requirement already satisfied: MarkupSafe==2.1.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 47)) (2.1.3)
Requirement already satisfied: matplotlib==3.7.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 48)) (3.7.2)
Requirement already satisfied: matplotlib-inline==0.1.6 in /home/jared/anaconda3/env
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s/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 49)) (0.1.6)
Requirement already satisfied: mistune==3.0.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 50)) (3.0.1)
Requirement already satisfied: nbclient==0.8.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 51)) (0.8.0)
Requirement already satisfied: nbconvert==7.7.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 52)) (7.7.4)
Requirement already satisfied: nbformat==5.9.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 53)) (5.9.2)
Requirement already satisfied: nest-asyncio==1.5.7 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 54)) (1.5.7)
Requirement already satisfied: notebook_shim==0.2.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 55)) (0.2.3)
Requirement already satisfied: numpy<1.24,>=1.22 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 56)) (1.23.5)
Requirement already satisfied: overrides==7.4.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 57)) (7.4.0)
Requirement already satisfied: packaging==23.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 58)) (23.1)
Requirement already satisfied: pandas<=1.5.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 59)) (1.5.3)
Requirement already satisfied: pandocfilters==1.5.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 60)) (1.5.0)
Requirement already satisfied: parso==0.8.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 61)) (0.8.3)
Requirement already satisfied: pexpect==4.8.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 62)) (4.8.0)
Requirement already satisfied: pickleshare==0.7.5 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 63)) (0.7.5)
Requirement already satisfied: Pillow==10.0.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 64)) (10.0.0)
Requirement already satisfied: platformdirs==3.10.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 65)) (3.10.0)
Requirement already satisfied: prometheus-client==0.17.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 66)) (0.17.1)
Requirement already satisfied: prompt-toolkit==3.0.39 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 67)) (3.0.39)
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Requirement already satisfied: psutil==5.9.5 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 68)) (5.9.5)

Requirement already satisfied: ptyprocess==0.7.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 69)) (0.7.0)

Requirement already satisfied: pure-eval==0.2.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 70)) (0.2.2)

Requirement already satisfied: pycparser==2.21 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 71)) (2.21)

Requirement already satisfied: Pygments==2.16.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 72)) (2.16.1)

Requirement already satisfied: pyparsing==3.0.9 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 73)) (3.0.9)

Requirement already satisfied: python-dateutil==2.8.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 74)) (2.8.2)

Requirement already satisfied: python-json-logger==2.0.7 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 75)) (2.0.7)

Requirement already satisfied: pytz==2023.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 76)) (2023.3)

Requirement already satisfied: PyYAML==6.0.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 77)) (6.0.1)

Requirement already satisfied: pyzmq<25 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 78)) (24.0.1)

Requirement already satisfied: referencing==0.30.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 79)) (0.30.2)

Requirement already satisfied: requests==2.31.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 80)) (2.31.0)

Requirement already satisfied: rfc3339-validator==0.1.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 81)) (0.1.4)

Requirement already satisfied: rfc3986-validator==0.1.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 82)) (0.1.1)

Requirement already satisfied: rpds-py==0.9.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 83)) (0.9.2)

Requirement already satisfied: scipy==1.11.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 84)) (1.11.2)

Requirement already satisfied: seaborn==0.12.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 85)) (0.12.2)

Requirement already satisfied: Send2Trash==1.8.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 86)) (1.8.2)

```
lab_requirements.txt (line 86)) (1.8.2)
Requirement already satisfied: six==1.16.0 in /home/jared/anaconda3/envs/CS6353/lib/
python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_re
quirements.txt (line 87)) (1.16.0)
Requirement already satisfied: sniffio==1.3.0 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 88)) (1.3.0)
Requirement already satisfied: soupsieve==2.4.1 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 89)) (2.4.1)
Requirement already satisfied: stack-data==0.6.2 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 90)) (0.6.2)
Requirement already satisfied: terminado==0.17.1 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 91)) (0.17.1)
Requirement already satisfied: tinycss2==1.2.1 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 92)) (1.2.1)
Requirement already satisfied: tornado<=6.3.2 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 93)) (6.3.2)
Requirement already satisfied: traitlets==5.9.0 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 94)) (5.9.0)
Requirement already satisfied: tzdata==2023.3 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 95)) (2023.3)
Requirement already satisfied: uri-template==1.3.0 in /home/jared/anaconda3/envs/CS6
353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/
colab_requirements.txt (line 96)) (1.3.0)
Requirement already satisfied: urllib3==2.0.4 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 97)) (2.0.4)
Requirement already satisfied: wcwidth==0.2.6 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 98)) (0.2.6)
Requirement already satisfied: webcolors==1.13 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 99)) (1.13)
Requirement already satisfied: webencodings==0.5.1 in /home/jared/anaconda3/envs/CS6
353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/
colab_requirements.txt (line 100)) (0.5.1)
Requirement already satisfied: websocket-client==1.6.2 in /home/jared/anaconda3/env
s/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignm
ent2/colab_requirements.txt (line 101)) (1.6.2)
Requirement already satisfied: exceptiongroup in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from anyio==3.7.1->-r /home/jared/CS-6353/Homeworks/ass
ignment2/colab_requirements.txt (line 1)) (1.3.0)
Requirement already satisfied: typing-extensions>=4.0.0 in /home/jared/anaconda3/env
s/CS6353/lib/python3.10/site-packages (from async-lru==2.0.4->-r /home/jared/CS-635
3/Homeworks/assignment2/colab_requirements.txt (line 7)) (4.15.0)
Requirement already satisfied: tomli in /home/jared/anaconda3/envs/CS6353/lib/python
3.10/site-packages (from jupyterlab==4.0.5->-r /home/jared/CS-6353/Homeworks/assignm
ent2/colab_requirements.txt (line 43)) (2.2.1)
Requirement already satisfied: ipython-genutils in /home/jared/anaconda3/envs/CS635
```

```
3/lib/python3.10/site-packages (from ipykernel<=5.5.6->-r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 28)) (0.2.0)
Requirement already satisfied: setuptools>=18.5 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from ipython<=7.34.0->-r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 29)) (78.1.1)
Requirement already satisfied: entrypoints in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from jupyter_client<8.0->-r /home/jared/CS-6353/Homeworks/assignment2/colab_requirements.txt (line 39)) (0.4)
```

Image features exercise

Complete and hand in this completed worksheet (including its outputs and any supporting code outside of the worksheet) with your assignment submission. For more details see the [assignments page](#) on the course website.

We have seen that we can achieve reasonable performance on an image classification task by training a linear classifier on the pixels of the input image. In this exercise we will show that we can improve our classification performance by training linear classifiers not on raw pixels but on features that are computed from the raw pixels.

All of your work for this exercise will be done in this notebook.

```
In [2]: from __future__ import print_function
import random
import numpy as np
from cs6353.data_utils import load_CIFAR10
import matplotlib.pyplot as plt

%matplotlib inline
plt.rcParams['figure.figsize'] = (10.0, 8.0) # set default size of plots
plt.rcParams['image.interpolation'] = 'nearest'
plt.rcParams['image.cmap'] = 'gray'

# for auto-reloading external modules
# see http://stackoverflow.com/questions/1907993/autoreload-of-modules-in-ipython
%load_ext autoreload
%autoreload 2
```

Load data

Similar to previous exercises, we will load CIFAR-10 data from disk.

```
In [3]: from cs6353.features import color_histogram_hsv, hog_feature

def get_CIFAR10_data(cifar10_dir='cs6353/datasets/cifar-10-batches-py', num_trainin
    # Load the raw CIFAR-10 data
    X_train, y_train, X_test, y_test = load_CIFAR10(cifar10_dir)

    # Subsample the data
```



```

    mask = list(range(num_training, num_training + num_validation))
    X_val = X_train[mask]
    y_val = y_train[mask]
    mask = list(range(num_training))
    X_train = X_train[mask]
    y_train = y_train[mask]
    mask = list(range(num_test))
    X_test = X_test[mask]
    y_test = y_test[mask]

    return X_train, y_train, X_val, y_val, X_test, y_test

# Cleaning up variables to prevent loading data multiple times (which may cause mem
try:
    del X_train, y_train
    del X_test, y_test
    print('Clear previously loaded data.')
except:
    pass

#####
# TODO: Change the path of the CIFAR-10 data directory correctly to           #
# the correct location                                                         #
# Default path is set to cs6353/datasets/cifar-10-batches-py'                 #
#####
cifar10_dir='cs6353/datasets/cifar-10-batches-py'
#####
#                                     END OF YOUR CODE                         #
#####

X_train, y_train, X_val, y_val, X_test, y_test = get_CIFAR10_data(cifar10_dir)

```

Extract Features

For each image we will compute a Histogram of Oriented Gradients (HOG) as well as a color histogram using the hue channel in HSV color space. We form our final feature vector for each image by concatenating the HOG and color histogram feature vectors.

Roughly speaking, HOG should capture the texture of the image while ignoring color information, and the color histogram represents the color of the input image while ignoring texture. As a result, we expect that using both together ought to work better than using either alone. Verifying this assumption would be a good thing to try for your interests.

The `hog_feature` and `color_histogram_hsv` functions both operate on a single image and return a feature vector for that image. The `extract_features` function takes a set of images and a list of feature functions and evaluates each feature function on each image, storing the results in a matrix where each column is the concatenation of all feature vectors for a single image.

```
In [4]: from cs6353.features import *
```



```
num_color_bins = 10 # Number of bins in the color histogram
feature_fns = [hog_feature, lambda img: color_histogram_hsv(img, nbin=num_color_bin
X_train_feats = extract_features(X_train, feature_fns, verbose=True)
X_val_feats = extract_features(X_val, feature_fns)
X_test_feats = extract_features(X_test, feature_fns)

# Preprocessing: Subtract the mean feature
mean_feat = np.mean(X_train_feats, axis=0, keepdims=True)
X_train_feats -= mean_feat
X_val_feats -= mean_feat
X_test_feats -= mean_feat

# Preprocessing: Divide by standard deviation. This ensures that each feature
# has roughly the same scale.
std_feat = np.std(X_train_feats, axis=0, keepdims=True)
X_train_feats /= std_feat
X_val_feats /= std_feat
X_test_feats /= std_feat

# Preprocessing: Add a bias dimension
X_train_feats = np.hstack([X_train_feats, np.ones((X_train_feats.shape[0], 1))])
X_val_feats = np.hstack([X_val_feats, np.ones((X_val_feats.shape[0], 1))])
X_test_feats = np.hstack([X_test_feats, np.ones((X_test_feats.shape[0], 1))])
```

Done extracting features for 1000 / 49000 images
Done extracting features for 2000 / 49000 images
Done extracting features for 3000 / 49000 images
Done extracting features for 4000 / 49000 images
Done extracting features for 5000 / 49000 images
Done extracting features for 6000 / 49000 images
Done extracting features for 7000 / 49000 images
Done extracting features for 8000 / 49000 images
Done extracting features for 9000 / 49000 images
Done extracting features for 10000 / 49000 images
Done extracting features for 11000 / 49000 images
Done extracting features for 12000 / 49000 images
Done extracting features for 13000 / 49000 images
Done extracting features for 14000 / 49000 images
Done extracting features for 15000 / 49000 images
Done extracting features for 16000 / 49000 images
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Done extracting features for 20000 / 49000 images
Done extracting features for 21000 / 49000 images
Done extracting features for 22000 / 49000 images
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Done extracting features for 44000 / 49000 images
Done extracting features for 45000 / 49000 images
Done extracting features for 46000 / 49000 images
Done extracting features for 47000 / 49000 images
Done extracting features for 48000 / 49000 images

Train SVM on features

Using the multiclass SVM code developed earlier in the assignment, train SVMs on top of the features extracted above; this should achieve better results than training SVMs directly on top of raw pixels.

```
In [5]: # Use the validation set to tune the learning rate and regularization strength

from cs6353.classifiers.linear_classifier import LinearSVM

# Learning_rates = [1e-9, 1e-8, 1e-7]
# regularization_strengths = [5e4, 5e5, 5e6]
learning_rates = np.logspace(-8,-3,7)
regularization_strengths = np.logspace(2,6,7)

results = {}
best_val = -1
best_svm = None

#####
# TODO:
# Use the validation set to set the learning rate and regularization strength.
# This should be identical to the validation that you did for the SVM; save
# the best trained classifier in best_svm. You might also want to play
# with different numbers of bins in the color histogram. If you are careful
# you should be able to get accuracy of near 0.44 on the validation set.
#####
for cur_lr in learning_rates:
    for r in regularization_strengths:
        curSVM = LinearSVM()
        curSVM.train(X_train_feats, y_train, cur_lr, r, num_iters=1000, batch_size=

        # Get training accuracy
        y_train_pred = curSVM.predict(X_train_feats)
        train_accuracy = np.mean(y_train == y_train_pred)

        # Get validation accuracy
        y_val_pred = curSVM.predict(X_val_feats)
        validation_accuracy = np.mean(y_val == y_val_pred)

        results[(cur_lr, r)] = (train_accuracy, validation_accuracy)
        if validation_accuracy > best_val:
            best_val = validation_accuracy
            best_svm = curSVM

#####
#                                     END OF YOUR CODE                                     #
#####

# Print out results.
for lr, reg in sorted(results):
    train_accuracy, val_accuracy = results[(lr, reg)]
    print('lr %e reg %e train accuracy: %f val accuracy: %f' % (
        lr, reg, train_accuracy, val_accuracy))
```

```
print('best validation accuracy achieved during cross-validation: %f' % best_val)
```

```
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/linear_svm.py:106: RuntimeWarning: overflow encountered in multiply
```

```
    dW += 2 * W * reg
```

```
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/linear_svm.py:78: RuntimeWarning: invalid value encountered in matmul
```

```
    scores = X @ W
```

```
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/linear_svm.py:80: RuntimeWarning: invalid value encountered in subtract
```

```
    margins = scores - correct_class_scores.reshape(-1,1) + 1
```

```

lr 1.000000e-08 reg 1.000000e+02 train accuracy: 0.085776 val accuracy: 0.090000
lr 1.000000e-08 reg 4.641589e+02 train accuracy: 0.088449 val accuracy: 0.081000
lr 1.000000e-08 reg 2.154435e+03 train accuracy: 0.143490 val accuracy: 0.155000
lr 1.000000e-08 reg 1.000000e+04 train accuracy: 0.108408 val accuracy: 0.124000
lr 1.000000e-08 reg 4.641589e+04 train accuracy: 0.097449 val accuracy: 0.099000
lr 1.000000e-08 reg 2.154435e+05 train accuracy: 0.114878 val accuracy: 0.120000
lr 1.000000e-08 reg 1.000000e+06 train accuracy: 0.412898 val accuracy: 0.404000
lr 6.812921e-08 reg 1.000000e+02 train accuracy: 0.139857 val accuracy: 0.147000
lr 6.812921e-08 reg 4.641589e+02 train accuracy: 0.100469 val accuracy: 0.092000
lr 6.812921e-08 reg 2.154435e+03 train accuracy: 0.090143 val accuracy: 0.091000
lr 6.812921e-08 reg 1.000000e+04 train accuracy: 0.096163 val accuracy: 0.093000
lr 6.812921e-08 reg 4.641589e+04 train accuracy: 0.384306 val accuracy: 0.379000
lr 6.812921e-08 reg 2.154435e+05 train accuracy: 0.415122 val accuracy: 0.413000
lr 6.812921e-08 reg 1.000000e+06 train accuracy: 0.401592 val accuracy: 0.391000
lr 4.641589e-07 reg 1.000000e+02 train accuracy: 0.132041 val accuracy: 0.149000
lr 4.641589e-07 reg 4.641589e+02 train accuracy: 0.144245 val accuracy: 0.159000
lr 4.641589e-07 reg 2.154435e+03 train accuracy: 0.243592 val accuracy: 0.225000
lr 4.641589e-07 reg 1.000000e+04 train accuracy: 0.414347 val accuracy: 0.417000
lr 4.641589e-07 reg 4.641589e+04 train accuracy: 0.412510 val accuracy: 0.414000
lr 4.641589e-07 reg 2.154435e+05 train accuracy: 0.402857 val accuracy: 0.399000
lr 4.641589e-07 reg 1.000000e+06 train accuracy: 0.324204 val accuracy: 0.336000
lr 3.162278e-06 reg 1.000000e+02 train accuracy: 0.367612 val accuracy: 0.362000
lr 3.162278e-06 reg 4.641589e+02 train accuracy: 0.406959 val accuracy: 0.397000
lr 3.162278e-06 reg 2.154435e+03 train accuracy: 0.413571 val accuracy: 0.407000
lr 3.162278e-06 reg 1.000000e+04 train accuracy: 0.415490 val accuracy: 0.412000
lr 3.162278e-06 reg 4.641589e+04 train accuracy: 0.371469 val accuracy: 0.390000
lr 3.162278e-06 reg 2.154435e+05 train accuracy: 0.275898 val accuracy: 0.259000
lr 3.162278e-06 reg 1.000000e+06 train accuracy: 0.100265 val accuracy: 0.087000
lr 2.154435e-05 reg 1.000000e+02 train accuracy: 0.416041 val accuracy: 0.417000
lr 2.154435e-05 reg 4.641589e+02 train accuracy: 0.416143 val accuracy: 0.418000
lr 2.154435e-05 reg 2.154435e+03 train accuracy: 0.409429 val accuracy: 0.420000
lr 2.154435e-05 reg 1.000000e+04 train accuracy: 0.361061 val accuracy: 0.356000
lr 2.154435e-05 reg 4.641589e+04 train accuracy: 0.104306 val accuracy: 0.099000
lr 2.154435e-05 reg 2.154435e+05 train accuracy: 0.100265 val accuracy: 0.087000
lr 2.154435e-05 reg 1.000000e+06 train accuracy: 0.100265 val accuracy: 0.087000
lr 1.467799e-04 reg 1.000000e+02 train accuracy: 0.410327 val accuracy: 0.414000
lr 1.467799e-04 reg 4.641589e+02 train accuracy: 0.402286 val accuracy: 0.424000
lr 1.467799e-04 reg 2.154435e+03 train accuracy: 0.366857 val accuracy: 0.362000
lr 1.467799e-04 reg 1.000000e+04 train accuracy: 0.088408 val accuracy: 0.094000
lr 1.467799e-04 reg 4.641589e+04 train accuracy: 0.100265 val accuracy: 0.087000
lr 1.467799e-04 reg 2.154435e+05 train accuracy: 0.100265 val accuracy: 0.087000
lr 1.467799e-04 reg 1.000000e+06 train accuracy: 0.100265 val accuracy: 0.087000
lr 1.000000e-03 reg 1.000000e+02 train accuracy: 0.396286 val accuracy: 0.384000
lr 1.000000e-03 reg 4.641589e+02 train accuracy: 0.348429 val accuracy: 0.333000
lr 1.000000e-03 reg 2.154435e+03 train accuracy: 0.100265 val accuracy: 0.087000
lr 1.000000e-03 reg 1.000000e+04 train accuracy: 0.100265 val accuracy: 0.087000
lr 1.000000e-03 reg 4.641589e+04 train accuracy: 0.100265 val accuracy: 0.087000
lr 1.000000e-03 reg 2.154435e+05 train accuracy: 0.100265 val accuracy: 0.087000
lr 1.000000e-03 reg 1.000000e+06 train accuracy: 0.100265 val accuracy: 0.087000
best validation accuracy achieved during cross-validation: 0.424000

```

```

In [6]: # Evaluate your trained SVM on the test set
y_test_pred = best_svm.predict(X_test_feats)
test_accuracy = np.mean(y_test == y_test_pred)
print(test_accuracy)

```


0.424

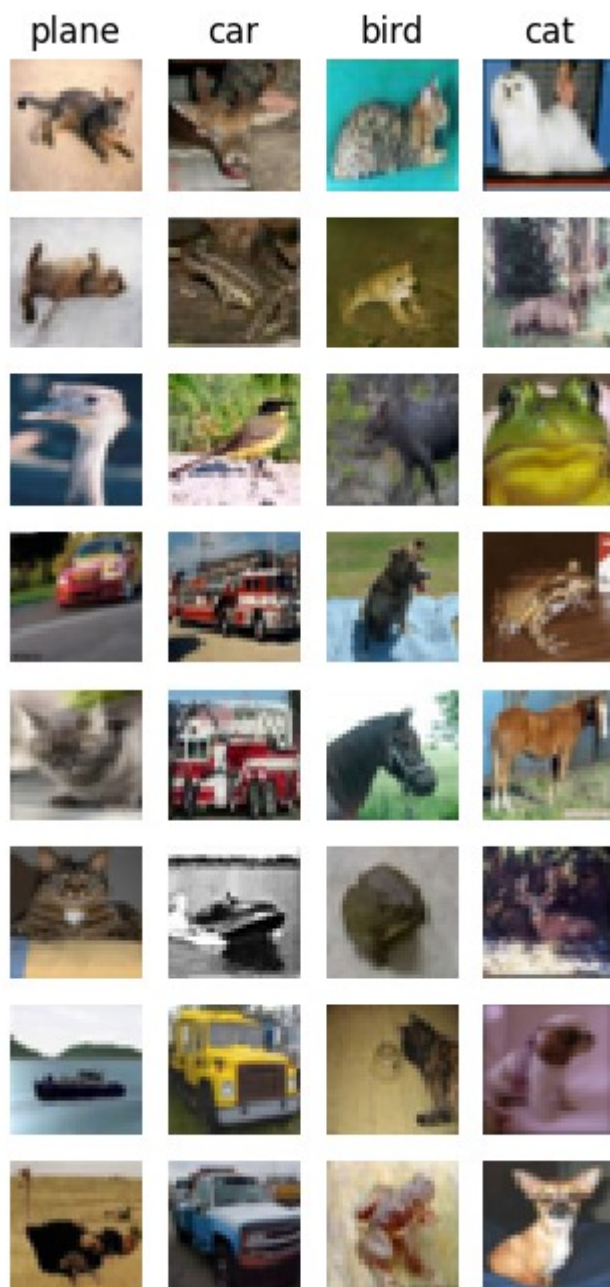
```
In [9]: # An important way to gain intuition about how an algorithm works is to
# visualize the mistakes that it makes. In this visualization, we show examples
# of images that are misclassified by our current system. The first column
# shows images that our system labeled as "plane" but whose true label is
# something other than "plane".

examples_per_class = 8
classes = ['plane', 'car', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', '
for cls, cls_name in enumerate(classes):
    idxs = np.where((y_test != cls) & (y_test_pred == cls))[0]
    idxs = np.random.choice(idxs, examples_per_class, replace=False)
    for i, idx in enumerate(idxs):
        plt.subplot(examples_per_class, len(classes), i * len(classes) + cls + 1)
        plt.imshow(X_test[idx].astype('uint8'))
        plt.axis('off')
        if i == 0:
            plt.title(cls_name)
plt.show()
```

```
-----
ValueError                                Traceback (most recent call last)
<ipython-input-9-1792925e61c9> in <cell line: 9>()
      9 for cls, cls_name in enumerate(classes):
     10     idxs = np.where((y_test != cls) & (y_test_pred == cls))[0]
--> 11     idxs = np.random.choice(idxs, examples_per_class, replace=False)
     12     for i, idx in enumerate(idxs):
     13         plt.subplot(examples_per_class, len(classes), i * len(classes) + cls
+ 1)

mtrand.pyx in numpy.random.mtrand.RandomState.choice()

ValueError: Cannot take a larger sample than population when 'replace=False'
```



Inline question 1:

Describe the misclassification results that you see. Do they make sense?

- Response: They seem to make sense in terms of our implementation. We remember that roughly speaking HOG represents texture and our histogram represents color distributions. The first thing that stood out to me is the airplane column. Color wise I can definitely see why that happened for a lot of them. The misclassified ones have a lot of the blue color in them. Another thing that stands out is the texture of the images. Every single image in the Airplane column has sharp points. If you squint that ostrich could very easily look like a plane. Each column has these color and texture patterns so it definitely makes sense what's going on here.
- Response: After some more hyper parameter tuning I noticed that this errored out. I

think this is because I don't have enough misclassifications for each class. The observations from above are still present.

Neural Network on image features

Earlier in this assignment we saw that training a two-layer neural network on raw pixels achieved better classification performance than linear classifiers on raw pixels. In this notebook we have seen that linear classifiers on image features outperform linear classifiers on raw pixels.

For completeness, we should also try training a neural network on image features. This approach should outperform all previous approaches: you should easily be able to achieve over 55% classification accuracy on the test set; our best model achieves about 60% classification accuracy.

```
In [10]: # Preprocessing: Remove the bias dimension
# Make sure to run this cell only ONCE
print(X_train_feats.shape)
X_train_feats = X_train_feats[:, :-1]
X_val_feats = X_val_feats[:, :-1]
X_test_feats = X_test_feats[:, :-1]

print(X_train_feats.shape)
```

```
(49000, 155)
```

```
(49000, 154)
```

```
In [11]: from cs6353.classifiers.neural_net import TwoLayerNet
#####
# TODO: Train a two-layer neural network on image features. You may want to #
# cross-validate various parameters as in previous sections. Store your best #
# model in the best_net variable. #
#####
# Your code
# Your code
best_net = None # store the best model into this
hidden_layer_sizes = [40, 80, 160, 320]
learning_rates = [1e-3, 1e-2, 1e-1, 1, 2]
num_epochs = [1000, 2000, 3000]
regularization_strengths = [0.001, 0.01, 0.01, 0.5, 10]

best_accuracy = -1
results = {}
best_args = ()

for cur_hidden_layer_size in hidden_layer_sizes:
    for cur_learning_rate in learning_rates:
        for cur_num_epoch in num_epochs:
            for cur_regularization_strength in regularization_strengths:
                input_size = X_train_feats.shape[1]
                num_classes = 10
```

```
net = TwoLayerNet(input_size, cur_hidden_layer_size, num_classes)

# Train the network
stats = net.train(X_train_feats, y_train, X_val_feats, y_val,
                  num_iters=cur_num_epoch, batch_size=200,
                  learning_rate=cur_learning_rate, learning_rate_decay=0.
                  reg=cur_regularization_strength, verbose=False)

# Predict on the validation set
val_acc = (net.predict(X_val_feats) == y_val).mean()

results[(cur_hidden_layer_size, cur_learning_rate, cur_num_epoch, c
if val_acc > best_accuracy:
    best_accuracy = val_acc
    best_args = (cur_hidden_layer_size, cur_learning_rate, cur_num_
    best_net = net

# Print out results.
for hidden_layer_size, learning_rate, num_epoch, regularization_strength in results:
    cur_val_acc = results[(hidden_layer_size, learning_rate, num_epoch, regularization_strength)
    print('hidden_layer_size: %d learning_rate: %f num_epoch: %d regularization_strength: %f val_acc: %f' %
          hidden_layer_size, learning_rate, num_epoch, regularization_strength, cur_val_acc)

# Print
best_hidden_layer_size, best_learning_rate, best_num_epoch, best_regularization_strength = best_args
print()
print("The best hyperparameters:")
print('hidden_layer_size: %d learning_rate: %f num_epoch: %d regularization_strength: %f val_acc: %f' %
      best_hidden_layer_size, best_learning_rate, best_num_epoch, best_regularization_strength, best_val_acc)

# hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 3000 regularization_strength: 0.000100 val_acc: 0.999900

#####
#                               END OF YOUR CODE                               #
#####
```

```
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/neural_net.py:103: RuntimeWarning: overflow encountered in exp
  loss_vec = -1 * np.log(np.exp(correct_class_scores) / np.sum(np.exp(scores), axis
= 1))
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/neural_net.py:103: RuntimeWarning: invalid value encountered in divide
  loss_vec = -1 * np.log(np.exp(correct_class_scores) / np.sum(np.exp(scores), axis
= 1))
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/neural_net.py:103: RuntimeWarning: divide by zero encountered in log
  loss_vec = -1 * np.log(np.exp(correct_class_scores) / np.sum(np.exp(scores), axis
= 1))
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/neural_net.py:120: RuntimeWarning: overflow encountered in exp
  s_js = np.exp(scores) / np.sum(np.exp(scores), axis = 1).reshape(-1,1)
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/neural_net.py:120: RuntimeWarning: invalid value encountered in divide
  s_js = np.exp(scores) / np.sum(np.exp(scores), axis = 1).reshape(-1,1)
```


hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 0.001000 validation_accuracy: 0.098000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 0.010000 validation_accuracy: 0.078000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 0.500000 validation_accuracy: 0.079000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 10.000000 validation_accuracy: 0.079000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 0.001000 validation_accuracy: 0.078000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 0.010000 validation_accuracy: 0.107000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 10.000000 validation_accuracy: 0.102000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 3000 regularization_strengt
h: 0.001000 validation_accuracy: 0.079000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 3000 regularization_strengt
h: 0.010000 validation_accuracy: 0.079000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 3000 regularization_strengt
h: 0.500000 validation_accuracy: 0.078000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 3000 regularization_strengt
h: 10.000000 validation_accuracy: 0.079000
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 0.001000 validation_accuracy: 0.079000
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 0.010000 validation_accuracy: 0.087000
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h: 0.500000 validation_accuracy: 0.119000
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 10.000000 validation_accuracy: 0.113000
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h: 0.001000 validation_accuracy: 0.179000
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h: 0.010000 validation_accuracy: 0.185000
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h: 0.500000 validation_accuracy: 0.079000
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 2000 regularization_strengt
h: 10.000000 validation_accuracy: 0.078000
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h: 0.001000 validation_accuracy: 0.216000
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h: 0.010000 validation_accuracy: 0.209000
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 3000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 3000 regularization_strengt
h: 10.000000 validation_accuracy: 0.078000
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hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
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hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 10.000000 validation_accuracy: 0.098000

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h: 0.001000 validation_accuracy: 0.523000

hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 0.010000 validation_accuracy: 0.510000

hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 0.500000 validation_accuracy: 0.098000

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h: 10.000000 validation_accuracy: 0.107000

hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 3000 regularization_strengt
h: 0.001000 validation_accuracy: 0.532000

hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 3000 regularization_strengt
h: 0.010000 validation_accuracy: 0.511000

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h: 0.500000 validation_accuracy: 0.098000

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h: 10.000000 validation_accuracy: 0.102000

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h: 0.001000 validation_accuracy: 0.536000

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h: 10.000000 validation_accuracy: 0.087000

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h: 0.001000 validation_accuracy: 0.550000

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h: 0.010000 validation_accuracy: 0.495000

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h: 0.500000 validation_accuracy: 0.098000

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h: 10.000000 validation_accuracy: 0.087000

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h: 0.001000 validation_accuracy: 0.558000

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h: 0.500000 validation_accuracy: 0.079000

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h: 10.000000 validation_accuracy: 0.087000

hidden_layer_size: 40 learning_rate: 2.000000 num_epoch: 1000 regularization_strengt
h: 0.001000 validation_accuracy: 0.531000

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h: 0.500000 validation_accuracy: 0.087000

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h: 10.000000 validation_accuracy: 0.087000

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h: 0.500000 validation_accuracy: 0.087000

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h: 10.000000 validation_accuracy: 0.087000

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h: 0.010000 validation_accuracy: 0.438000

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h: 0.500000 validation_accuracy: 0.087000

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h: 10.000000 validation_accuracy: 0.087000

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h: 0.001000 validation_accuracy: 0.098000

hidden_layer_size: 80 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 0.010000 validation_accuracy: 0.078000

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h: 0.500000 validation_accuracy: 0.087000

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h: 10.000000 validation_accuracy: 0.078000

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h: 0.001000 validation_accuracy: 0.079000

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h: 0.010000 validation_accuracy: 0.113000

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h: 0.500000 validation_accuracy: 0.079000

hidden_layer_size: 80 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 10.000000 validation_accuracy: 0.079000

hidden_layer_size: 80 learning_rate: 0.001000 num_epoch: 3000 regularization_strengt
h: 0.001000 validation_accuracy: 0.079000

hidden_layer_size: 80 learning_rate: 0.001000 num_epoch: 3000 regularization_strengt
h: 0.010000 validation_accuracy: 0.087000

hidden_layer_size: 80 learning_rate: 0.001000 num_epoch: 3000 regularization_strengt
h: 0.500000 validation_accuracy: 0.105000

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h: 10.000000 validation_accuracy: 0.078000

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h: 0.001000 validation_accuracy: 0.107000

hidden_layer_size: 80 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 0.010000 validation_accuracy: 0.079000

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h: 0.500000 validation_accuracy: 0.087000

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h: 0.010000 validation_accuracy: 0.178000

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h: 0.500000 validation_accuracy: 0.078000

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h: 10.000000 validation_accuracy: 0.079000

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h: 0.500000 validation_accuracy: 0.112000

hidden_layer_size: 80 learning_rate: 0.010000 num_epoch: 3000 regularization_strengt
h: 10.000000 validation_accuracy: 0.112000

hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 0.001000 validation_accuracy: 0.508000

hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 0.010000 validation_accuracy: 0.481000

hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 0.500000 validation_accuracy: 0.098000

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h: 10.000000 validation_accuracy: 0.105000

hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 0.001000 validation_accuracy: 0.536000

hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 0.010000 validation_accuracy: 0.515000

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h: 0.500000 validation_accuracy: 0.079000

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h: 10.000000 validation_accuracy: 0.079000

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h: 0.001000 validation_accuracy: 0.551000

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h: 0.010000 validation_accuracy: 0.505000

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h: 0.500000 validation_accuracy: 0.098000

hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 3000 regularization_strengt
h: 10.000000 validation_accuracy: 0.078000

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h: 0.500000 validation_accuracy: 0.078000

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h: 10.000000 validation_accuracy: 0.087000

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h: 0.010000 validation_accuracy: 0.492000

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h: 0.500000 validation_accuracy: 0.112000

hidden_layer_size: 80 learning_rate: 1.000000 num_epoch: 2000 regularization_strengt
h: 10.000000 validation_accuracy: 0.087000

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h: 0.010000 validation_accuracy: 0.507000

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h: 0.500000 validation_accuracy: 0.098000

hidden_layer_size: 80 learning_rate: 1.000000 num_epoch: 3000 regularization_strengt
h: 10.000000 validation_accuracy: 0.087000

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h: 0.001000 validation_accuracy: 0.531000

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h: 10.000000 validation_accuracy: 0.087000

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hidden_layer_size: 80 learning_rate: 2.000000 num_epoch: 2000 regularization_strengt
h: 0.010000 validation_accuracy: 0.435000
hidden_layer_size: 80 learning_rate: 2.000000 num_epoch: 2000 regularization_strengt
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hidden_layer_size: 80 learning_rate: 2.000000 num_epoch: 2000 regularization_strengt
h: 10.000000 validation_accuracy: 0.087000
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h: 0.500000 validation_accuracy: 0.087000
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h: 10.000000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 1000 regularization_streng
th: 0.001000 validation_accuracy: 0.079000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 1000 regularization_streng
th: 0.010000 validation_accuracy: 0.107000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 1000 regularization_streng
th: 0.500000 validation_accuracy: 0.078000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 1000 regularization_streng
th: 10.000000 validation_accuracy: 0.079000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 2000 regularization_streng
th: 0.001000 validation_accuracy: 0.078000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 2000 regularization_streng
th: 0.010000 validation_accuracy: 0.078000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 2000 regularization_streng
th: 0.500000 validation_accuracy: 0.079000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 2000 regularization_streng
th: 10.000000 validation_accuracy: 0.078000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 3000 regularization_streng
th: 0.001000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 3000 regularization_streng
th: 0.010000 validation_accuracy: 0.119000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 3000 regularization_streng
th: 0.500000 validation_accuracy: 0.079000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 3000 regularization_streng
th: 10.000000 validation_accuracy: 0.087000
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th: 0.500000 validation_accuracy: 0.079000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 1000 regularization_streng
th: 10.000000 validation_accuracy: 0.079000
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th: 0.010000 validation_accuracy: 0.189000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 2000 regularization_streng
th: 0.500000 validation_accuracy: 0.079000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 2000 regularization_streng
th: 10.000000 validation_accuracy: 0.078000

hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 3000 regularization_strength: 0.001000 validation_accuracy: 0.241000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 3000 regularization_strength: 0.010000 validation_accuracy: 0.233000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 3000 regularization_strength: 0.500000 validation_accuracy: 0.078000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 3000 regularization_strength: 10.000000 validation_accuracy: 0.079000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 1000 regularization_strength: 0.001000 validation_accuracy: 0.510000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 1000 regularization_strength: 0.010000 validation_accuracy: 0.498000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 1000 regularization_strength: 0.500000 validation_accuracy: 0.078000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 1000 regularization_strength: 10.000000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 2000 regularization_strength: 0.001000 validation_accuracy: 0.536000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 2000 regularization_strength: 0.010000 validation_accuracy: 0.515000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 2000 regularization_strength: 0.500000 validation_accuracy: 0.107000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 2000 regularization_strength: 10.000000 validation_accuracy: 0.102000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 3000 regularization_strength: 0.001000 validation_accuracy: 0.553000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 3000 regularization_strength: 0.010000 validation_accuracy: 0.510000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 3000 regularization_strength: 0.500000 validation_accuracy: 0.098000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 3000 regularization_strength: 10.000000 validation_accuracy: 0.078000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 1000 regularization_strength: 0.001000 validation_accuracy: 0.562000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 1000 regularization_strength: 0.010000 validation_accuracy: 0.447000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 1000 regularization_strength: 0.500000 validation_accuracy: 0.112000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 1000 regularization_strength: 10.000000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 2000 regularization_strength: 0.001000 validation_accuracy: 0.573000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 2000 regularization_strength: 0.010000 validation_accuracy: 0.520000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 2000 regularization_strength: 0.500000 validation_accuracy: 0.105000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 2000 regularization_strength: 10.000000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 3000 regularization_strength: 0.001000 validation_accuracy: 0.577000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 3000 regularization_strength: 0.010000 validation_accuracy: 0.502000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 3000 regularization_strength: 0.500000 validation_accuracy: 0.079000
hidden_layer_size: 160 learning_rate: 1.000000 num_epoch: 3000 regularization_strength: 10.000000 validation_accuracy: 0.087000

hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 1000 regularization_strength: 0.001000 validation_accuracy: 0.545000
hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 1000 regularization_strength: 0.010000 validation_accuracy: 0.471000
hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 1000 regularization_strength: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 1000 regularization_strength: 10.000000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 2000 regularization_strength: 0.001000 validation_accuracy: 0.550000
hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 2000 regularization_strength: 0.010000 validation_accuracy: 0.439000
hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 2000 regularization_strength: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 2000 regularization_strength: 10.000000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 3000 regularization_strength: 0.001000 validation_accuracy: 0.573000
hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 3000 regularization_strength: 0.010000 validation_accuracy: 0.497000
hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 3000 regularization_strength: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 2.000000 num_epoch: 3000 regularization_strength: 10.000000 validation_accuracy: 0.087000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 1000 regularization_strength: 0.001000 validation_accuracy: 0.112000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 1000 regularization_strength: 0.010000 validation_accuracy: 0.102000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 1000 regularization_strength: 0.500000 validation_accuracy: 0.079000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 1000 regularization_strength: 10.000000 validation_accuracy: 0.087000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 2000 regularization_strength: 0.001000 validation_accuracy: 0.078000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 2000 regularization_strength: 0.010000 validation_accuracy: 0.078000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 2000 regularization_strength: 0.500000 validation_accuracy: 0.079000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 2000 regularization_strength: 10.000000 validation_accuracy: 0.079000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 3000 regularization_strength: 0.001000 validation_accuracy: 0.079000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 3000 regularization_strength: 0.010000 validation_accuracy: 0.087000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 3000 regularization_strength: 0.500000 validation_accuracy: 0.098000
hidden_layer_size: 320 learning_rate: 0.001000 num_epoch: 3000 regularization_strength: 10.000000 validation_accuracy: 0.079000
hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 1000 regularization_strength: 0.001000 validation_accuracy: 0.105000
hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 1000 regularization_strength: 0.010000 validation_accuracy: 0.079000
hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 1000 regularization_strength: 0.500000 validation_accuracy: 0.078000
hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 1000 regularization_strength: 10.000000 validation_accuracy: 0.078000

hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 2000 regularization_strength: 0.001000 validation_accuracy: 0.177000
hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 2000 regularization_strength: 0.010000 validation_accuracy: 0.186000
hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 2000 regularization_strength: 0.500000 validation_accuracy: 0.079000
hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 2000 regularization_strength: 10.000000 validation_accuracy: 0.098000
hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 3000 regularization_strength: 0.001000 validation_accuracy: 0.265000
hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 3000 regularization_strength: 0.010000 validation_accuracy: 0.240000
hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 3000 regularization_strength: 0.500000 validation_accuracy: 0.078000
hidden_layer_size: 320 learning_rate: 0.010000 num_epoch: 3000 regularization_strength: 10.000000 validation_accuracy: 0.098000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 1000 regularization_strength: 0.001000 validation_accuracy: 0.518000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 1000 regularization_strength: 0.010000 validation_accuracy: 0.497000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 1000 regularization_strength: 0.500000 validation_accuracy: 0.119000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 1000 regularization_strength: 10.000000 validation_accuracy: 0.087000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 2000 regularization_strength: 0.001000 validation_accuracy: 0.529000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 2000 regularization_strength: 0.010000 validation_accuracy: 0.498000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 2000 regularization_strength: 0.500000 validation_accuracy: 0.107000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 2000 regularization_strength: 10.000000 validation_accuracy: 0.112000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 3000 regularization_strength: 0.001000 validation_accuracy: 0.558000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 3000 regularization_strength: 0.010000 validation_accuracy: 0.505000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 3000 regularization_strength: 0.500000 validation_accuracy: 0.112000
hidden_layer_size: 320 learning_rate: 0.100000 num_epoch: 3000 regularization_strength: 10.000000 validation_accuracy: 0.113000
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 1000 regularization_strength: 0.001000 validation_accuracy: 0.563000
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 1000 regularization_strength: 0.010000 validation_accuracy: 0.459000
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 1000 regularization_strength: 0.500000 validation_accuracy: 0.078000
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 1000 regularization_strength: 10.000000 validation_accuracy: 0.087000
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 2000 regularization_strength: 0.001000 validation_accuracy: 0.560000
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 2000 regularization_strength: 0.010000 validation_accuracy: 0.473000
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 2000 regularization_strength: 0.500000 validation_accuracy: 0.078000
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 2000 regularization_strength: 10.000000 validation_accuracy: 0.087000

```
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 3000 regularization_streng  
th: 0.001000 validation_accuracy: 0.583000  
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 3000 regularization_streng  
th: 0.010000 validation_accuracy: 0.495000  
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 3000 regularization_streng  
th: 0.500000 validation_accuracy: 0.112000  
hidden_layer_size: 320 learning_rate: 1.000000 num_epoch: 3000 regularization_streng  
th: 10.000000 validation_accuracy: 0.087000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 1000 regularization_streng  
th: 0.001000 validation_accuracy: 0.552000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 1000 regularization_streng  
th: 0.010000 validation_accuracy: 0.441000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 1000 regularization_streng  
th: 0.500000 validation_accuracy: 0.087000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 1000 regularization_streng  
th: 10.000000 validation_accuracy: 0.087000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 2000 regularization_streng  
th: 0.001000 validation_accuracy: 0.564000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 2000 regularization_streng  
th: 0.010000 validation_accuracy: 0.470000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 2000 regularization_streng  
th: 0.500000 validation_accuracy: 0.087000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 2000 regularization_streng  
th: 10.000000 validation_accuracy: 0.087000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 3000 regularization_streng  
th: 0.001000 validation_accuracy: 0.570000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 3000 regularization_streng  
th: 0.010000 validation_accuracy: 0.467000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 3000 regularization_streng  
th: 0.500000 validation_accuracy: 0.087000  
hidden_layer_size: 320 learning_rate: 2.000000 num_epoch: 3000 regularization_streng  
th: 10.000000 validation_accuracy: 0.087000
```

The best hyperparameters:

```
hidden_layer_size: 80 learning_rate: 1.000000 num_epoch: 3000 regularization_strengt  
h: 0.001000 validation_accuracy: 0.584000
```

In [12]: *# Run your best neural net classifier on the test set. You should be able
to get more than 55% accuracy.*

```
test_acc = (best_net.predict(X_test_feats) == y_test).mean()  
print(test_acc)
```

0.564